# Increased Cholesterol Efflux Potential of Sera From ApoA-$I_{Milano}$ Carriers and Transgenic Mice

Guido Franceschini, Laura Calabresi, Giulia Chiesa, Cinzia Parolini, Cesare R. Sirtori, Monica Canavesi, Franco Bernini

***Abstract***—The ability of HDL to remove cholesterol from peripheral cells and drive it to the liver for excretion is believed to explain most of the strong inverse correlation between plasma HDL cholesterol levels and coronary heart disease. Carriers of the ApoA-$I_{Milano}$ (A-$I_M$) mutant have a severe hypoalphalipoproteinemia but are not at increased risk for premature of coronary heart disease. To explain this apparent paradox, we compared the capacity of serum from A-$I_M$ and control subjects to extract cholesterol from Fu5AH cells. Because the A-$I_M$ carriers are all heterozygotes for the mutation, we also compared the cholesterol efflux capacity of serum from transgenic mice expressing A-$I_M$ or wild-type ApoA-I (A-$I_{WT}$), in the absence of murine ApoA-I. In the whole series of human or mouse sera, cholesterol efflux was significantly correlated with several HDL-related parameters; after adjustment for concomitant variables, the only parameter that remained significantly correlated with cholesterol efflux was the serum ApoA-I concentration ($r^2$=0.85 in humans and 0.84 in mice). The same was true when samples from control subjects, A-$I_M$ carriers, A-$I_{WT}$ or A-$I_M$ mice were analyzed separately. Cholesterol efflux to sera from the A-$I_M$ carriers was only reduced slightly compared with control sera (25.0±4.2% versus 30.4±3.3%), although there was a large reduction (−45%) in the serum ApoA-I concentration in the former. Cholesterol efflux was also lower to sera from A-$I_M$ than A-$I_{WT}$ mice (15.6±3.8% versus 30.1±7.1%), but less than expected from the 70% reduction in serum ApoA-I concentration. A relative efflux potential of serum was calculated in each group as the slope of the regression line fitting cholesterol efflux to ApoA-I concentrations. Therefore, the relative efflux potential reflects the relative efficiency of ApoA-I in determining cell cholesterol efflux. The relative efflux potential of mouse and human sera was in the following order: A-$I_M$ mice>A-$I_M$ carriers>A-$I_{WT}$ mice=control subjects, suggesting a gene–dosage effect of the A-$I_M$ mutation on the efficiency of serum to extract cholesterol from cells. The high efficiency of A-$I_M$-containing HDL for cell cholesterol uptake would result in an improved reverse cholesterol transport in the A-$I_M$ carriers, possibly explaining the low susceptibility to atherosclerosis development. **(*Arterioscler Thromb Vasc Biol*. 1999;19:1257-1262.)**

**Key Words:** reverse cholesterol transport ■ ApoA-I ■ cholesterol efflux ■ HDL ■ transgenic mice

Numerous case–control and prospective studies have shown a strong inverse relation between plasma HDL cholesterol (HDL cholesterol) concentrations and the risk of coronary heart disease.[1] The mechanism(s) underlying this negative association are not fully understood, and several biological explanations have been proposed.[2] Most prominent seems the function of HDL as vehicle of cholesterol in the reverse cholesterol transport (RCT),[3,4] the process by which excess cholesterol in peripheral tissues, including the arterial wall, is transported to the liver for excretion.

The first step in RCT is the efflux of unesterified cholesterol from peripheral cells to either free apolipoproteins[5] or lipoprotein acceptors[6] present in the extracellular space. Several serum factors, including the acceptor structure/composition, and the concentration of lecithin:cholesterol acyltransferase (LCAT), cholesteryl ester transfer protein (CETP), and other lipoproteins are known to modulate cell cholesterol efflux.[7] Therefore, the use of whole serum instead of isolated lipoprotein fractions would provide an ideal tool to evaluate the individual capacity to promote cell cholesterol efflux.[8] This approach has the advantage that all of the potential factors affecting efflux are present in the experimental setting, and therefore seems particularly suitable to correlate the efficiency of the earliest step in RCT to atherosclerosis susceptibility in subjects,[9,10] or animal models at different risks of cardiovascular disease.[11,12]

ApoA-$I_{Milano}$ (A-$I_M$) is the first described mutant of human apolipoproteins.[13] Thirty-eight heterozygous carriers have been identified, up to now,[14] who represent the largest group of individuals with low plasma HDL levels because of a single defect in the ApoA-I gene. The plasma concentration of major factors involved in RCT, ie, HDL, ApoA-I, LpA-I,

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LCAT, and CETP is remarkably reduced in A-I$_M$ carriers compared with controls.[15–17] The severe hypoalphalipoproteinemia and the partial LCAT and CETP deficiencies suggest a defective RCT, but the carriers do not suffer from premature coronary heart disease.[14] To investigate the mechanism(s) behind this apparent paradox, we compared in the present study the cholesterol efflux potential of sera from A-I$_M$ carriers and control subjects. Because of the limited number of carriers and the difficulty in discerning the impact of the mutation in these heterozygous subjects, we also evaluated the cholesterol efflux potential of sera from transgenic mice expressing the human A-I$_M$ variant or wild-type ApoA-I, in the absence of murine ApoA-I. A-I$_M$ transgenic mice share with human carriers low plasma HDL levels, increased triglycerides, defective cholesterol esterification, and structural abnormalities in HDL particles,[18,19] and therefore are a suitable model for studying the impact of the mutation in a homozygous condition.

## Methods

### Subjects

Nineteen adult heterozygous A-I$_M$ carriers (9 females and 10 males), belonging to the previously described A-I$_M$ kindred,[14] volunteered for this study. Nineteen age- and sex-matched noncarriers were selected among close relatives in the same families. All participating subjects were healthy, were taking no medications, and consumed a typical Mediterranean diet. All subjects were fully informed of the methods and end points of the study, which were approved by the Internal Review Board.

After an overnight fast, blood was collected both into empty plastic tubes and into tubes containing $Na_2$-EDTA (final concentration, 1 mg/mL). Serum and plasma were prepared by low-speed centrifugation at 4°C. Serum aliquots were added with $Na_2$-EDTA (1 mg/mL) and solid NaBr (final concentration, 5.1 mol/L) and kept at 4°C for HDL subfraction analysis by rate-zonal ultracentrifugation.[20] Serum aliquots for cholesterol efflux were immediately frozen and stored at −80°C until assayed.

### Transgenic Mice

The generation of transgenic mice expressing human ApoA-I (A-I$_{WT}$) and the A-I$_M$ variant has been previously described.[18] These mice express A-I$_{WT}$ or A-I$_M$ together with human ApoA-II, and do not express murine ApoA-I because of gene targeting.[21] Twenty-one A-I$_M$ mice and 21 A-I$_{WT}$ mice, of similar age (16 to 24 weeks) and of both sexes, were used for the cholesterol efflux experiments.

Blood was collected after an overnight fast from the retroorbital plexus. Serum was prepared by low-speed centrifugation at 4°C and assayed on the same day for human apolipoprotein and lipid/lipoprotein levels. Serum aliquots for cholesterol efflux were immediately frozen and stored at −80°C until assayed.

### Cell Cholesterol Efflux

The cholesterol efflux potential of serum was assayed as described by de la Llera Moya et al,[8] by incubating diluted serum with [$^3$H]cholesterol-labeled Fu5AH rat hepatoma cells for 4 hours at 37°C. Cells were seeded in Corning 24-well (15.5 mm/well) plates, using 20 000 cells per well, and grown in DMEM with 5% FCS for 2 days. Lipids were radiolabeled by adding 2 μCi/mL [1,2-$^3$H]cholesterol (Amersham) to 25% FCS in DMEM. Cells were grown in the presence of radiolabeled cholesterol for 2 additional days to obtain confluent monolayers. The labeling medium was replaced with DMEM containing 1% essential fatty acid–free albumin for 18 to 20 hours to allow equilibration of the label. Cells were then washed 2 times with PBS and incubated for 4 hours with control medium or serum diluted in DMEM with essential fatty acid–free albumin 1%. All efflux assays were performed by using the same dilution (5%) for mouse and human sera. At the end of this period, the medium was removed, collected into tubes, and centrifuged for 5 minutes at 2000 rpm to remove any floating cells. An aliquot of the medium was then counted for [$^3$H]cholesterol radioactivity (Formula 989, Packard). Cellular lipids were extracted with 2-propanol by overnight incubation at room temperature and radioactivity was measured in an aliquot of the extract (Insta-Fluor, Packard). Cholesterol efflux was calculated as the percentage of total label in each well released to the medium. Individual efflux values were calculated as averages of 3 determinations in different wells, normalized to the cholesterol efflux obtained with a pool of normolipidemic sera tested in each experiment.

### Analyses

Serum total and unesterified cholesterol (TC and UC), triglycerides (TG), and phospholipid (PL) levels were determined with standard enzymatic techniques by using a Roche diagnostics Cobas autoanalyzer. The CE mass was calculated as (TC−UC)×1.68. The protein (P) content in lipoprotein fractions was determined by the method of Lowry et al.[22] The lipoprotein surface/core ratio was calculated as (FC+PL+P)/(TG+CE). HDL cholesterol levels in human sera were measured after precipitation of the ApoB-containing lipoproteins by dextran sulfate–$MgCl_2$.[23] HDL cholesterol was measured in mouse sera after precipitation of ApoB-containing lipoproteins with polyethylene glycol (20%, wt/vol) in 0.2 mol/L glycine (pH 10).[24] ApoA-I, ApoA-II, and ApoB levels were determined by immunoturbidimetry,[25] using the Cobas analyzer with commercially available polyclonal antibodies (Boehringer Mannheim). The anti-A-I antibody recognizes all forms of A-I$_M$ (monomer, homodimer, and heterodimer) as wild-type ApoA-I[26]; therefore, the ApoA-I concentration determined in the sera of A-I$_M$ carriers, who are heterozygotes for the mutation, is the sum of mutant and wild-type ApoA-I, and the ApoA-I concentration in the sera of A-I$_M$ mice, who are homozygotes for the mutation, is the concentration of mutant ApoA-I.

Human plasma lipoproteins were separated by sequential ultracentrifugation,[27] using a Beckman TL 100 ultracentrifuge equipped with a TL 100.3 rotor (Beckman Instruments). HDL subfractions were isolated by rate-zonal ultracentrifugation in a swinging bucket rotor[20]; 2 fractions, designated as $HDL_2$ and $HDL_3$, were collected, and the total cholesterol content measured by enzyme methods.

### Statistical Analyses

Quantitative variables are expressed as mean±SD values. Differences among the groups were evaluated by 1-way ANOVA, with post hoc evaluation by the Neuman–Keuls test. Statistical significance was defined as $P<0.05$. Simple and multivariate regression analyses were performed and the significance of the correlations was determined by the $F$ parameter. In the forward stepwise regression, the independent parameters were included 1 at a time starting with the parameter that had the highest correlation with the dependent variable, fractional cholesterol efflux. Additional parameters were included only if a significant increase in goodness of fit was achieved. Logarithmic transformation was performed on individual data when values were not normally distributed.

## Results

### Cholesterol Efflux to Human Sera

The fasting serum lipid and lipoprotein concentrations in the examined A-I$_M$ carriers and control subjects are given in Table 1. The A-I$_M$ carriers had significantly lower serum total and LDL cholesterol, and phospholipid concentrations than controls; although there was a trend for A-I$_M$ carriers to have greater serum triglycerides and VLDL cholesterol levels, the difference in average values between carriers and controls did not reach statistical significance ($P=0.15$ and 0.16, respectively). The unesterified/esterified cholesterol ratio was significantly higher in the A-I$_M$ carriers (0.45±0.09 versus 0.34±0.05), mostly because of the lower plasma LCAT concentration.[16] As expected,[15] serum HDL cholesterol, $HDL_2$ cholesterol, $HDL_3$ cholesterol, HDL phospholipid, ApoA-I, and ApoA-II levels were much lower in the A-I$_M$

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**TABLE 1. Serum Lipid/Lipoprotein Levels in A-$I_M$ Carriers and Controls**

| | A-$I_M$ Carriers | Controls |
|---|---|---|
| Sex (F/M) | 9/10 | 9/10 |
| Age (years) | 44.7±18.4 | 45.4±16.4 |
| BMI (kg/m$^2$) | 24.2±3.2 | 24.0±2.6 |
| Total cholesterol (mg/dL) | 181.1±42.8* | 231.2±35.4 |
| VLDL cholesterol (mg/dL) | 36.9±19.2 | 28.8±14.7 |
| LDL cholesterol (mg/dL) | 126.2±30.6* | 154.2±28.0 |
| HDL cholesterol (mg/dL) | 18.1±10.5* | 48.2±11.8 |
| $HDL_2$ cholesterol (mg/dL) | 3.2±2.0* | 12.8±6.6 |
| $HDL_3$ cholesterol (mg/dL) | 15.1±8.5* | 35.4±7.9 |
| Phospholipids (mg/dL) | 222.2±62.9* | 259.8±40.4 |
| HDL phospholipids (mg/dL) | 69.9±32.2* | 104.9±34.0 |
| Triglycerides (mg/dL) | 184.3±96.2 | 144.0±73.5 |
| ApoA-I (mg/dL) | 76.8±33.4* | 138.8±28.9 |
| ApoA-II (mg/dL) | 17.5±9.0* | 37.4±6.4 |
| ApoB (mg/dL) | 120.4±39.9 | 126.0±28.3 |
| HDL phospholipids/ApoA-I | 0.95±0.27 | 0.75±0.18 |

Data are expressed as mean±SD values.
*Significantly different from controls.

carriers than controls. The ApoA-I/ApoA-II ratio was significantly higher in the A-$I_M$ carriers than in controls (4.65±1.20 versus 3.70±0.65), consistent with the reported preferential reduction of LpA-I:A-II particles in the former.[17]

HDL particles from A-$I_M$ carriers were enriched in triglycerides and phospholipids, and depleted in cholesteryl esters compared with control HDL (Table 2); the surface/core ratio was significantly higher in A-$I_M$ HDL than in control HDL (4.29±1.26 versus 3.25±0.49), indicative of a prevalence of small particles in the serum of A-$I_M$ carriers.[28]

Cholesterol efflux data, expressed as percent efflux from Fu5AH cells during a 4-hour incubation, obtained with sera from A-$I_M$ carriers and controls, are shown in Figure 1. The average efflux value was 18% lower in A-$I_M$ carriers (25.0±4.2%) than in controls (30.4±3.3; $P<0.001$). Both female and male subjects participated in this study; a significantly higher cholesterol efflux to sera from female than male subjects was found among A-$I_M$ carriers (27.0±4.0% versus 23.1±3.6%) but not controls (30.6±3.2% versus 30.1±3.6%). The trend toward higher efflux with A-$I_M$ female serum was paralleled by a trend for higher HDL cholesterol levels in female than male serum (20.6±9.9 and 15.9±10.9 mg/dL, respectively).

**TABLE 2. Serum Levels of HDL Components in A-$I_M$ Carriers and Controls**

| | A-$I_M$ Carriers | Controls |
|---|---|---|
| Unesterified cholesterol | 4.6±2.3* (2.1±0.4) | 7.0±3.3 (1.8±0.6) |
| Cholesterol esters | 22.7±14.1* (10.2±3.1*) | 69.2±16.2 (18.4±1.6) |
| Triglycerides | 19.4±8.2 (9.7±3.4*) | 19.5±6.7 (5.4±2.2) |
| Phospholipids | 69.9±32.2* (33.4±5.0*) | 104.9±34.0 (27.3±4.0) |
| Proteins | 94.3±40.0* (44.6±5.6) | 176.3±33.7 (47.1±3.7) |

Data are mean±SD values and are expressed as milligrams per deciliter (percentages within parentheses).
*Significantly different from controls.

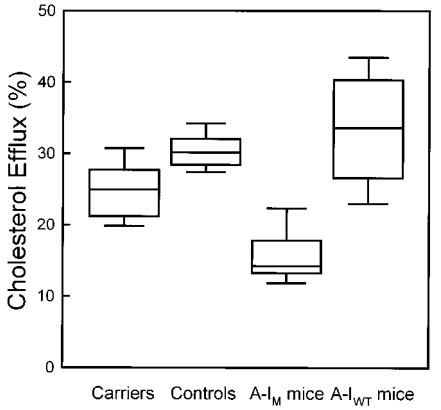



**Figure 1.** Box plot of the cholesterol efflux from Fu5AH cells incubated with serum from A-$I_M$ carriers, control subjects, and transgenic mice expressing the A-$I_M$ mutant (A-$I_M$ mice) or wild-type ApoA-I (A-$I_{WT}$ mice). Data represent the percentages of cholesterol released to the medium during a 4-hour incubation. The plot displays median values with the 25th and 75th percentiles; capped bars indicate the 10th and 90th percentiles.

Cholesterol efflux in the whole series of sera correlated with several serum parameters, most of which are related to HDL (Table 3). Stepwise regression analysis was then performed to identify which of the correlated parameters best predicted cholesterol efflux to serum. In the whole series, the serum ApoA-I concentration was the strongest predictor of cholesterol efflux ($r^2=0.85$); HDL triglycerides, HDL phospholipids, and HDL unesterified cholesterol made small additions to this correlation, the 4 variables explaining 90% of the variation in cholesterol efflux. A separate stepwise analysis was also performed with data from control and A-$I_M$ samples. Serum ApoA-I concentration was again the strongest predictor of cholesterol efflux to control sera ($r^2=0.63$), with serum esterified cholesterol adding significantly to this

**TABLE 3. Univariate Correlations Between Cholesterol Efflux and Lipid/Lipoprotein Concentrations in the Examined Sera**

| Independent Variable | Whole Series | A-$I_M$ Carriers | Controls |
|---|---|---|---|
| Total cholesterol | 0.590 | *0.298* | 0.520 |
| Esterified cholesterol | 0.659 | *0.366* | 0.516 |
| Unesterified/esterified cholesterol | −0.514 | *−0.257* | *−0.029* |
| Phospholipids | 0.430 | *0.154* | 0.615 |
| ApoA-I | 0.874 | 0.928 | 0.791 |
| ApoA-II | 0.793 | 0.707 | 0.697 |
| LDL cholesterol | 0.384 | *0.198* | *0.333* |
| HDL cholesterol | 0.752 | 0.805 | 0.519 |
| HDL-unesterified cholesterol | 0.606 | 0.779 | *0.305* |
| HDL esterified cholesterol | 0.739 | 0.790 | 0.531 |
| HDL unesterified/esterified cholesterol | −0.328 | −0.406 | *0.083* |
| HDL triglycerides | 0.495 | 0.653 | *0.040* |
| HDL phospholipids | 0.744 | 0.789 | 0.476 |
| HDL proteins | 0.832 | 0.866 | 0.560 |
| $HDL_2$ cholesterol | 0.570 | 0.681 | *0.323* |
| $HDL_3$ cholesterol | 0.773 | 0.813 | 0.513 |

Parameters that are not significantly correlated with cholesterol efflux in the whole series are not reported. Nonsignificant correlation coefficients in separate groups of A-$I_M$ carriers and controls are italicized.

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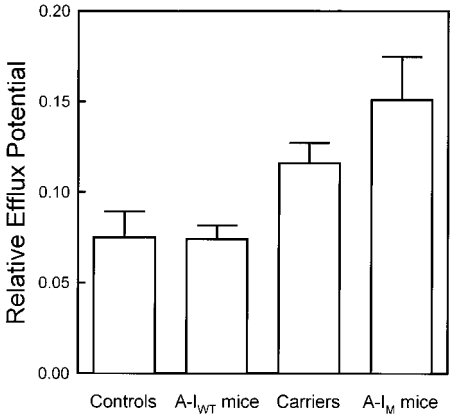



**Figure 2.** REP of sera from control subjects, A-I$_M$ carriers, and transgenic mice expressing wild-type ApoA-I (A-I$_{WT}$ mice) or the A-I$_M$ mutant (A-I$_M$ mice). The REP was calculated as the slope of the regression lines fitting cholesterol efflux to ApoA-I concentrations in the various groups. REP±SEM values are shown; differences among various groups are significant, with $P=0.0027$.

prediction ($r^2=0.72$). In the A-I$_M$ group, serum ApoA-I was the strongest univariate predictor of cholesterol efflux ($r^2=0.86$); HDL phospholipids and the HDL unesterified/esterified cholesterol made small additions to this correlation, the 3 variables explaining 93% of the variation in cholesterol efflux.

Because the serum ApoA-I concentration is the best predictor of cholesterol efflux, a relative efflux potential (REP) was calculated as the slope of the regression line fitting cholesterol efflux to ApoA-I concentrations[11] for A-I$_M$ and control subjects, separately. Therefore, REP reflects the relative efficiency of ApoA-I in determining cell cholesterol efflux. The REP was 50% higher in A-I$_M$ carriers (0.116) than controls (0.075) (Figure 2), indicating that the same relative variation in ApoA-I concentration causes a higher cholesterol efflux to the serum of A-I$_M$ carriers than of control subjects. This difference in REP explains the relatively mild reduction in cholesterol efflux to A-I$_M$ than control sera, despite the severe ApoA-I reduction in A-I$_M$ sera.

### Cholesterol Efflux to Mouse Sera

One of the major problems we encountered in evaluating the impact of the A-I$_M$ mutation on lipoprotein metabolism in the carriers is that they are all heterozygotes for the mutation, carrying both wild-type and mutant ApoA-I.[14] To resolve this issue, transgenic mice were generated that express the A-I$_M$ mutant (A-I$_M$ mice) in the absence of murine ApoA-I.[18] We then compared the capacity of the serum from these mice to extract cholesterol from cells with that of serum from mice expressing wild-type human ApoA-I (A-I$_{WT}$ mice). The serum lipid/lipoprotein levels in A-I$_M$ and A-I$_{WT}$ transgenic mice are reported in Table 4. The A-I$_M$ mice had significantly higher serum triglycerides and lower total cholesterol, HDL cholesterol, HDL phospholipids, and ApoA-I concentrations than A-I$_{WT}$ mice.[18] Therefore, the expression of the A-I$_M$ mutation in a transgenic mouse reproduces the major lipid/lipoprotein abnormalities observed in human carriers, providing a unique model to investigate the metabolic effects of the A-I$_M$ mutation in a homozygous state.

The cholesterol efflux to mouse serum was strongly positively correlated with the serum ApoA-I concentration in the whole series of samples ($r=0.92$), and both in the A-I$_M$ and A-I$_{WT}$ groups ($r=0.77$ and 0.83, respectively), as found in experiments with human sera. Indeed, the serum ApoA-I concentration was the only parameter that remained significantly correlated with cholesterol efflux after adjustment for concomitant variables. Cholesterol efflux to serum was significantly lower in A-I$_M$ than A-I$_{WT}$ mice (Figure 1 and Table 4). The serum REP was calculated as the slope of the regression lines fitting cholesterol efflux to ApoA-I concentrations, as done for human samples. The REP was 2-fold higher in A-I$_M$ than A-I$_{WT}$ mice (0.151 versus 0.074) (Figure 2), indicating that for the same relative variation in serum ApoA-I concentration, the percent cholesterol efflux is higher with A-I$_M$ than A-I$_{WT}$ mice sera, as found in human subjects.

It is noteworthy that the REP was similar for the sera from A-I$_{WT}$ mice and from control subjects (Figure 2), indicating that the efficiency of human ApoA-I in determining cholesterol efflux is the same when the protein is expressed in either human or murine background. Indeed a strong positive correlation ($r=0.87$) was found between cholesterol efflux and serum ApoA-I concentration, when data from control subjects and A-I$_{WT}$ mice were analyzed together.

**TABLE 4. Serum Lipid/Lipoprotein Levels and Cholesterol Efflux to Whole Serum in A-I$_M$ and A-I$_{WT}$ Transgenic Mice**

| | A-I$_M$ Mice | A-I$_{WT}$ Mice |
|---|---|---|
| Total cholesterol (mg/dL) | 33.3±13.0* | 105.5±25.1 |
| Triglycerides (mg/dL) | 85.0±17.5* | 60.5±10.5 |
| HDL cholesterol (mg/dL) | 17.2±6.3* | 96.7±26.9 |
| HDL phospholipids (mg/dL) | 56.9±21.8* | 155.8±44.3 |
| ApoA-I (mg/dL) | 55.3±19.3* | 183.8±93.1 |
| HDL phospholipids/ApoA-I | 1.05±0.25 | 1.04±0.47 |
| Cholesterol efflux (%) | 15.6±3.8* | 33.2±7.7 |

Data are expressed as mean±SD values.
*Significantly different from A-I$_{WT}$ mice.

## Discussion

Cholesterol efflux from peripheral cells to HDL acceptors is the first step in RCT, the pathway believed to explain the protective role of HDL against the development of vascular lesions.[7] To investigate whether the low susceptibility of the A-I$_M$ carriers to atherosclerotic vascular disease may be explained by improved cholesterol efflux capacity, we examined the ability of serum from these subjects and from transgenic mice expressing the A-I$_M$ mutant to extract cholesterol from cultured cells. The present results indicate that in a series of samples from humans and mice expressing wild-type or mutant ApoA-I, the concentration of ApoA-I is the largest predictor of the ability of serum to extract cholesterol from cells. They also provide indirect evidence of a similar efficiency of human ApoA-I in promoting cell cholesterol efflux when the protein is expressed in humans or mice, and of an improved efficiency of the A-I$_M$ mutant versus wild-type ApoA-I in determining cell cholesterol efflux both in humans and mice.

The correlation analyses on the whole series of human data and on data from separate groups indicate that cholesterol efflux to serum is positively correlated with several HDL-related parameters, confirming that HDL particles are the

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major factors responsible for cholesterol efflux from cells to serum.[8] By multivariate correlation analysis, the ApoA-I concentration in serum was the largest predictor of cholesterol efflux. A strong positive correlation between serum ApoA-I levels and cholesterol efflux was also found when data from 2 animal species expressing human ApoA-I, as control subjects and A-$I_{WT}$ mice, were combined. Thus, it follows that human ApoA-I determines the cholesterol efflux potential of serum, even in species with a widely different genetic background.

Recent cholesterol efflux studies with human samples, and with sera from transgenic mice and rats expressing human ApoA-I, suggest that the major serum factor involved in the regulation of cell cholesterol efflux is the HDL phospholipids content.[11,29–31] Phospholipids may facilitate the interaction of HDL acceptors with cells through the scavenger receptor BI,[32,33] therefore improving cholesterol desorption from the cell membrane. This is particularly true for Fu5AH cells, which display a high level of scavenger receptor BI expression.[32] Indeed, in the present study, HDL phospholipid was strongly correlated with efflux, and contributed to the variability in cholesterol efflux among the various serum samples. However, the serum ApoA-I concentration was the strongest predictor of cholesterol efflux to serum. The reasons for this discrepancy are not immediately clear, and may relate to the different level of ApoA-I expression in the various transgenic lines. In transgenic animals expressing high levels of human ApoA-I, as those used in previous studies,[11,29] the cholesterol efflux to serum became less efficient as the concentration of ApoA-I increased, because of a marked decrease in the HDL phospholipid/ApoA-I ratio, and to the appearance of poorly effective lipid-free ApoA-I in serum.[11,29] This is not the case in the present study. The A-$I_M$ and A-$I_{WT}$ mice have low-normal serum ApoA-I levels, no lipid-free ApoA-I in serum,[18,31] and a similar HDL phospholipid/ApoA-I ratio. The distinct physicochemical properties of acceptor particles containing wild-type or mutant ApoA-I[18,28,34,35] may also contribute to the discrepancy between the present and previous findings, by affecting the interaction of HDL with scavenger receptor BI.

A major observation in this study was that, although there was a large reduction in serum ApoA-I and HDL concentrations, the cholesterol efflux to sera from the A-$I_M$ carriers was only slightly reduced compared with control sera. The explanation for this apparent paradox came from the estimation of the efficiency of A-$I_M$ and A-$I_{WT}$ in determining cell cholesterol efflux, obtained by calculating the serum REP as the slope of the linear regression line between cholesterol efflux and serum ApoA-I concentration. The REP, which provides an indirect estimation of the relative efficiency of acceptor particles to extract cholesterol from cells,[11] was higher in A-$I_M$ carriers than control subjects, suggesting a relative abundance of more efficient acceptors in the sera from A-$I_M$ carriers. Previous extensive studies on the characterization of HDL particles in these subjects demonstrate that A-$I_M$ HDL are smaller in size than control HDL,[28,34] a characteristic that has been associated with improved efficiency for cell cholesterol uptake. Moreover, efflux studies with reconstituted HDL show that particles containing a recombinant form of the disulfide-linked A-$I_M$ dimer are more efficient than A-$I_{WT}$-containing particles in removing cholesterol from cultured cells,[36] possibly because of the unique conformation of the mutant ApoA-I on the surface of HDL.[35] A direct effect of the mutation on the efficiency of HDL acceptors in cell cholesterol uptake is demonstrated by the control subjects=A-$I_{WT}$ mice<A-$I_M$ carriers<A-$I_M$ mice gradient in the REP (Figure 2). Indeed, after adjustment for concomitant variables, a highly significant ($P=0.009$) gene–dosage effect of the A-$I_M$ mutation on REP was found.

Cholesterol efflux from cells to serum acceptors is only the first step in RCT, a pathway that involves many other processes.[3,4] Therefore, variations in the serum capacity to extract cholesterol from cells would only partially contribute to the efficiency of RCT in vivo, and to the individual cardiovascular risk. Nevertheless, it is noteworthy that the A-$I_M$ carriers, who are not at increased risk despite the severe hypoalphalipoproteinemia, display an improved serum capacity for cell cholesterol uptake. This finding enhances our understanding of the impact of the A-$I_M$ mutation on RCT, and provides an explanation for the apparent protection of A-$I_M$ carriers against atherosclerotic vascular disease. It also supports the concept that cholesterol efflux is a major determinant of RCT in vivo,[37] and may contribute significantly to the determination of individual cardiovascular risk.

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## References

1. Gordon DJ, Rifkind BM. High density lipoprotein: the clinical implications of recent studies. *N Engl J Med*. 1989;321:1311–1316.
2. Badimon JJ, Fuster V, Badimon L. Role of high density lipoproteins in the regression of atherosclerosis. *Circulation*. 1992;86:III-86–III-94.
3. Glomset JA. The plasma lecithin:cholesterol acyltransferase reaction. *J Lipid Res*. 1968;9:155–162.
4. Fielding CJ, Fielding PE. Molecular physiology of reverse cholesterol transport. *J Lipid Res*. 1995;36:211–228.
5. Oram JF, Yokoyama S. Apolipoprotein-mediated removal of cellular cholesterol and phospholipids. *J Lipid Res*. 1996;37:2473–2491.
6. Rodrigueza WV, Williams KJ, Rothblat GH, Phillips MC. Remodeling and shuttling. Mechanisms for the synergistic effects between different acceptor particles in the mobilization of cellular cholesterol. *Arterioscler Thromb Vasc Biol*. 1997;17:383–393.
7. Franceschini G, Werba JP, Calabresi L. Drug control of reverse cholesterol transport. *Pharmacol Ther*. 1994;61:289–324.
8. de la Llera Moya M, Atger V, Paul JL, Fournier N, Moatti N, Giral P, Friday KE, Rothblat GH. A cell culture system for screening human serum for ability to promote cellular cholesterol efflux. Relations between serum components and efflux, esterification, and transfer. *Arterioscler Thromb*. 1994;14:1056–1065.
9. Syvanne M, Castro G, Dengremont C, De Geitere C, Jauhiainen M, Ehnholm C, Michelagnoli S, Franceschini G, Kahri J, Taskinen MR. Cholesterol efflux from Fu5AH hepatoma cells induced by plasma of subjects with or without coronary artery disease and non-insulin-dependent diabetes: importance of LpA-I:A-II particles and phospholipid transfer protein. *Atherosclerosis*. 1996;127:245–253.
10. Elkhalil L, Majd Z, Bakir R, Perez-Mendez O, Castro G, Poulain P, Lacroix B, Duhal N, Fruchart JC, Luc G. Fish-eye disease: structural and in vivo metabolic abnormalities of high-density lipoproteins. *Metabolism*. 1997;46:474–483.
11. Atger V, de la Llera Moya M, Bamberger M, Francone O, Cosgrove P, Tall AR, Walsh A, Moatti N, Rothblat G. Cholesterol efflux potential of sera from mice expressing human cholesteryl ester transfer protein and/or human apolipoprotein AI. *J Clin Invest*. 1995;96:2613–2622.
12. Castro G, Nihoul LP, Dengremont C, De Geitere C, Delfly B, Tailleux A, Fievet C, Duverger N, Denefle P, Fruchart JC, Rubin EM. Cholesterol efflux, lecithin-cholesterol acyltransferase activity, and pre-beta particle formation by serum from human apolipoprotein A-I and apolipoprotein A-I/

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apolipoprotein A-II transgenic mice consistent with the latter being less effective for reverse cholesterol transport. *Biochemistry*. 1997;36:2243–2249.
13. Franceschini G, Sirtori CR, Capurso A, Weisgraber KH, Mahley RW. A-I$_{Milano}$ apoprotein: decreased high density lipoprotein cholesterol levels with significant lipoprotein modifications and without clinical atherosclerosis in an Italian family. *J Clin Invest*. 1980;66:892–900.
14. Gualandri V, Franceschini G, Sirtori CR, Gianfranceschi G, Orsini GB, Cerrone A, Menotti A. A-I$_{Milano}$ apoprotein: identification of the complete kindred and evidence of a dominant genetic transmission. *Am J Hum Genet*. 1985;37:1083–1097.
15. Franceschini G, Sirtori CR, Bosisio E, Gualandri V, Orsini GB, Mogavero AM, Capurso A. Relationship of the phenotypic expression of the A-I Milano apoprotein with plasma lipid and lipoprotein patterns. *Atherosclerosis*. 1985;58:159–174.
16. Franceschini G, Baio M, Calabresi L, Sirtori CR, Cheung MC. Apolipoprotein A-I$_{Milano:}$ partial lecithin:cholesterol acyltransferase deficiency due to low levels of a functional enzyme. *Biochim Biophys Acta*. 1990; 1043:1–6.
17. Bekaert ED, Alaupovic P, Knight Gibson C, Franceschini G, Sirtori CR. Apolipoprotein A-I$_{Milano:}$ sex-related differences in the concentration and composition of apoA-I- and apoB-containing lipoprotein particles. *J Lipid Res*. 1993;34:111–123.
18. Chiesa G, Stoltzfus LJ, Michelagnoli S, Bielicki JK, Santi M, Forte TM, Sirtori CR, Franceschini G, Rubin EM. Elevated triglycerides and low HDL cholesterol in transgenic mice expressing human apolipoprotein A-I$_{Milano}$. *Atherosclerosis*. 1998;136:139–146.
19. Bielicki JK, Forte TM, McCall MR, Stoltzfus LJ, Chiesa G, Sirtori CR, Franceschini G, Rubin EM. High density lipoprotein particle size restriction in apolipoprotein A-I$_{Milano}$ transgenic mice. *J Lipid Res*. 1997; 38:2314–2321.
20. Franceschini G, Tosi C, Moreno Y, Sirtori CR. Effects of storage on the distribution of high density lipoprotein subfractions in human sera. *J Lipid Res*. 1985;26:1368–1373.
21. Li H, Reddick RL, Maeda N. Lack of apoA-I is not associated with increased susceptibility to atherosclerosis in mice. *Arterioscler Thromb*. 1993;13:1814–1821.
22. Lowry OH, Rosebrough NJ, Farr AL, Randall RJ. Protein measurement with the Folin phenol reagent. *J Biol Chem*. 1951;193:265–275.
23. Warnick GR, Benderson J, Albers JJ. Dextran sulfate precipitation procedure for quantitation of high density lipoproteins. *Clin Chem*. 1982;28: 1379–1388.
24. Schultz JR, Gong EL, McCall MR, Nichols AV, Clift SM, Rubin EM. Expression of human apolipoprotein A-II and its effect on high density lipoproteins in transgenic mice. *J Biol Chem*. 1992;267:21630–21636.
25. Rifai N, King ME. Immunoturbidimetric assays of apolipoproteins A, AI, AII and B in serum. *Clin Chem*. 1986;32:957–961.
26. Weisgraber KH, Bersot TP, Mahley RW, Franceschini G, Sirtori CR. A-I Milano apoprotein: isolation and characterization of a cysteine-containing variant of the A-I apoprotein from human high density lipoproteins. *J Clin Invest*. 1980;66:901–907.
27. Havel RJ, Eder HA, Bragdon JH. The distribution and chemical composition of ultracentrifugally separated lipoproteins in human serum. *J Clin Invest*. 1955;34:1345–1354.
28. Franceschini G, Calabresi L, Tosi C, Sirtori CR, Fragiacomo GC, Noseda G, Gong EL, Blanche PJ, Nichols AV. Apolipoprotein A-I$_{Milano:}$ correlation between lipoprotein subclass distribution and triglyceridemia. *Arteriosclerosis*. 1987;7:426–435.
29. Fournier N, de la Llera Moya M, Burkey BF, Swaney JB, Paterniti JR Jr, Moatti N, Atger V, Rothblat GH. Role of HDL phospholipid in efflux of cell cholesterol to whole serum: studies with human apoA-I transgenic rats. *J Lipid Res*. 1996;37:1704–1711.
30. Jian B, de la Llera-Moya M, Royer L, Rothblat G, Francone O, Swaney JB. Modification of the cholesterol efflux properties of human serum by enrichment with phospholipid. *J Lipid Res*. 1997;38:734–744.
31. Chiesa G, Parolini C, Canavesi M, Sirtori CR, Fumagalli R, Franceschini G, Bernini F. Human apolipoproteins A-I and A-II in cell cholesterol efflux: studies with transgenic mice. *Arterioscler Thromb Vasc Biol*. 1998;18:1417–1423.
32. Ji Y, Jian B, Wang N, Sun Y, Moya ML, Phillips MC, Rothblat GH, Swaney JB, Tall AR. Scavenger receptor BI promotes high density lipoprotein-mediated cellular cholesterol efflux. *J Biol Chem*. 1997;272: 20982–20985.
33. Jian B, de la Llera-Moya M, Ji Y, Wang N, Phillips MC, Swaney JB, Tall AR, Rothblat GH. Scavenger receptor class B type I as a mediator of cellular cholesterol efflux to lipoproteins and phospholipid acceptors. *J Biol Chem*. 1998;273:5599–5606.
34. Franceschini G, Frosi triglycerides, Manzoni C, Gianfranceschi G, Sirtori CR. High density lipoprotein-3 heterogeneity in subjects with the apoAI$_{Milano}$ variant. *J Biol Chem*. 1982;257:9926–9930.
35. Calabresi L, Vecchio G, Frigerio F, Vavassori L, Sirtori CR, Franceschini G. Reconstituted high-density lipoproteins with a disulfide-linked apolipoprotein A-I dimer: evidence for restricted particle size heterogeneity. *Biochemistry*. 1997;36:12428–12433.
36. Franceschini G, Calabresi L, Sirtori CR, Bernini F, Jonas A, Taramelli D, Vecchio G. Structure-function relationships in apolipoprotein A-I: insights from the A-I$_{Milano}$ mutation. In: Jacotot B, Mathé D, Fruchart JC, eds. *Atherosclerosis XI*. Singapore: Elsevier Science Pte Ltd; 1998: 1129–1134.
37. Stein O, Dabach Y, Hollander G, Ben-Naim M, Halperin G, Breslow JL, Stein Y. Delayed loss of cholesterol from a localized lipoprotein depot in apolipoprotein A-I-deficient mice. *Proc Natl Acad Sci U S A*. 1997;94: 9820–9824.

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